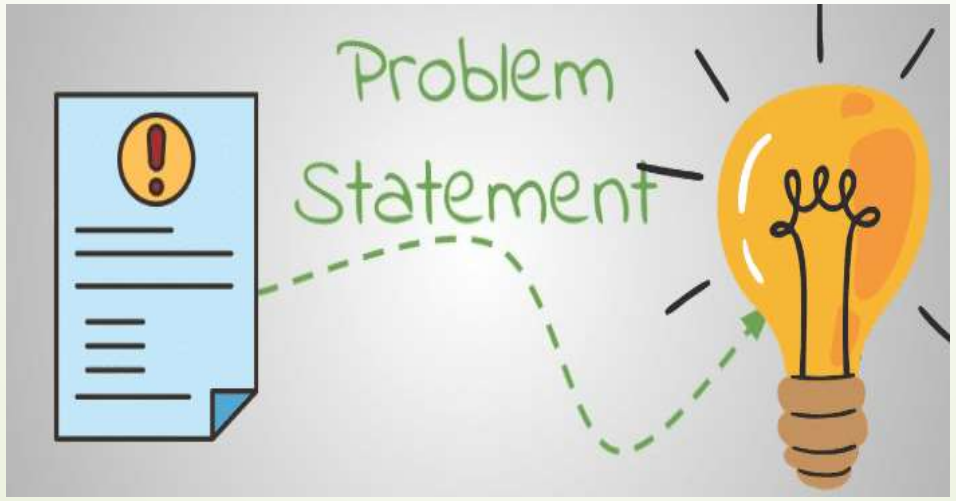
ANALYSIS ON ROAD ACCIDENTS IN COIMBATORE (2022-24)

INTRODUCTION

- Road accidents are a **global** problem, resulting in millions of deaths and injuries every year.
- According to the World Health Organization, approximately **1.35 million** people lose their lives to road traffic accidents annually, and **between 20 and 50 million** more suffer non-fatal injuries.
- These accidents are not only a leading cause of death but also a significant source of physical and psychological trauma for victims and their families.
- The number and quantity of vehicles on roads has increased due to **technological** and **economic development** in recent years.
- As a result of this increase, traffic has been **one of the 16** most important parts of our daily lives as people spend more time in traffic thereby forcing drivers and other road users to face a **higher risk** of traffic accident.

PROBLEM STATEMENT

- Coimbatore, a rapidly growing city in Tamil Nadu, India, is experiencing a **significant increase** in road accidents, raising concerns about public safety and the efficiency of its transportation system.
- The problem is **multi-faceted**, involving factors such as poor road infrastructure, increasing vehicle density, varying driving behaviors, and insufficient traffic management.



DATA DESCRIPTION

The dataset contains **secondary data** on road accidents that occurred in Coimbatore, India, collected from the Coimbatore Police Station. The data provides insights into the **frequency, causes, and consequences** of road accidents in Coimbatore .

Data Source- Coimbatore Police Station

- **Type of Data** : Secondary Data
- **Sample Size** : **1555**
- **Mode of Collection** : Coimbatore Police Station

OBJECTIVES

1.Identify Accident Hotspots: Pinpoint specific locations within the Coimbatore city that experience a high frequency of road accidents.

2.Analyze Contributing Factors: Determine the primary causes of accidents, such as poor road conditions, driver behaviors like Hit and drive, Rash driving and drunk & drive.

3.Assess Accident Severity: Evaluate the severity of accidents, including the number of injuries and fatalities, to understand the impact on public safety with respect to the vehicle type.

4.Evaluate Demographic Factors:Study the involvement of different demographic groups, such as age, gender, and occupation, in accidents.

5.Understand Temporal Patterns: Analyze the timing of accidents to identify peak times or seasons when accidents are more likely to occur.

6.Forecasting : To predict the number of accidents that happen in the future.

TOOLS USED

- Identify Accident Hotspots:

  Data visualization in POWER BI

- Understand Temporal Patterns:

  Trend Line in Excel

- Analyze Contributing Factors:

  Descriptive Statistics + Visualization in Power BI

- Assess Accident Severity:

  Chi-Square Test in SPSS + Visualization in Power BI

- Evaluate Demographic Factors:

  Chi-Square Test in SPSS + Visualization in Power BI

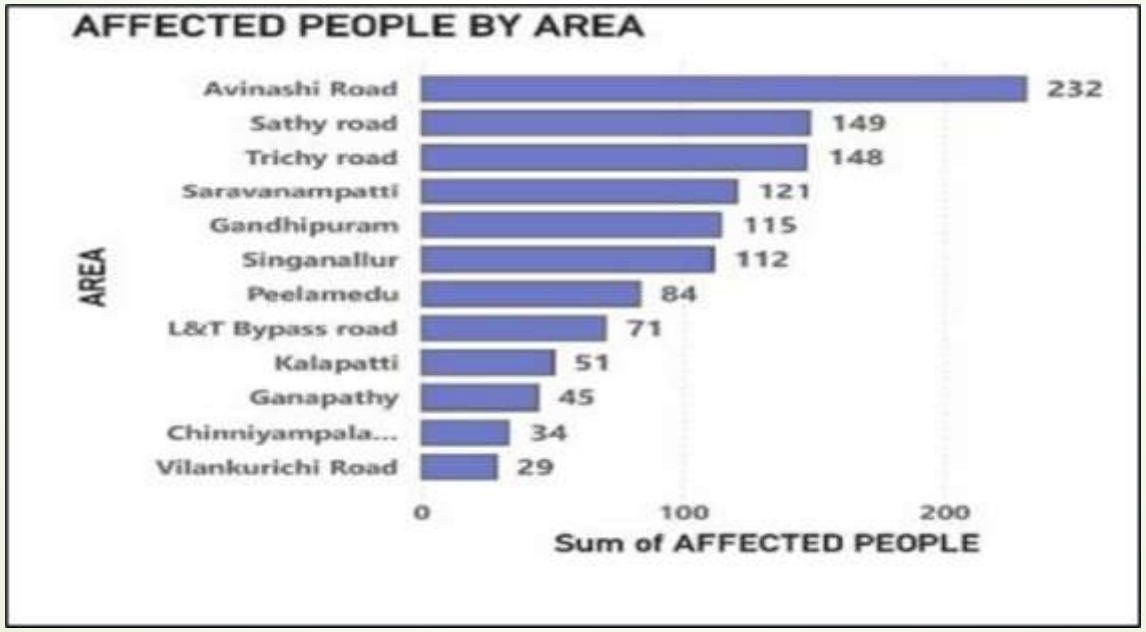
- Forecasting:

  Exponential Smoothing (Forecast sheet) in Excel

RESULTS AND INTERPRETATION:

1) Identifying Accident Hotspots:

- **AIM:** To Pinpoint specific locations within the Coimbatore city that experience a high frequency of road accidents.
- **TOOLS USED:** Descriptive statistics (excel), visualization(Power BI)
- **RESULTS:**



INTERPRETATION: The highest number of accidents took place in the **Avinashi Road (232)** and the least number of accidents took place in the **Vilankurichi Road (29).**

2) Analyze Contributing Factors:

- **AIM:** Determine the primary causes of accidents, such as poor road conditions, traffic congestion, driver behavior, weather conditions, and vehicle types.
- **TOOLS USED :** Descriptive statistics , visualization (excel)
- **RESULTS:**

Descriptive statistics:

Mean	92.15789
Standard Error	57.88627
Median	2
Mode	1
Standard Deviation	252.3204
Sample Variance	63665.58
Kurtosis	6.42001
Skewness	2.775174
Range	810
Minimum	1
Maximum	811
Sum	1751
Count	19

CAUSE	TOTAL ACCIDENTS	TOTAL DEATHS
Drunk and drive	*74*	*6*
Hit and run	*8*	*0*
Rash and negligent driving	*1643*	*212*
Self hit	*28*	*13*
Total	*1753*	*231*

INTERPRETATION:

1) Causes, such as **rash and negligent driving or drunk and drive**, result in a high number of accidents, most other causes, like car door opened or chest pain, are less frequent.

2) The high standard deviation (**252.32**) and variance further indicate that the number of accidents caused by different factors **varies widely**.

3) The positive skewness (**2.78**) and kurtosis **(6.42)** suggest that the distribution is heavily influenced by a **few extreme values**, meaning that while many causes result in only a small number of accidents, a few (like rash driving) are responsible for a disproportionately high number.

3) Assess Accident Severity

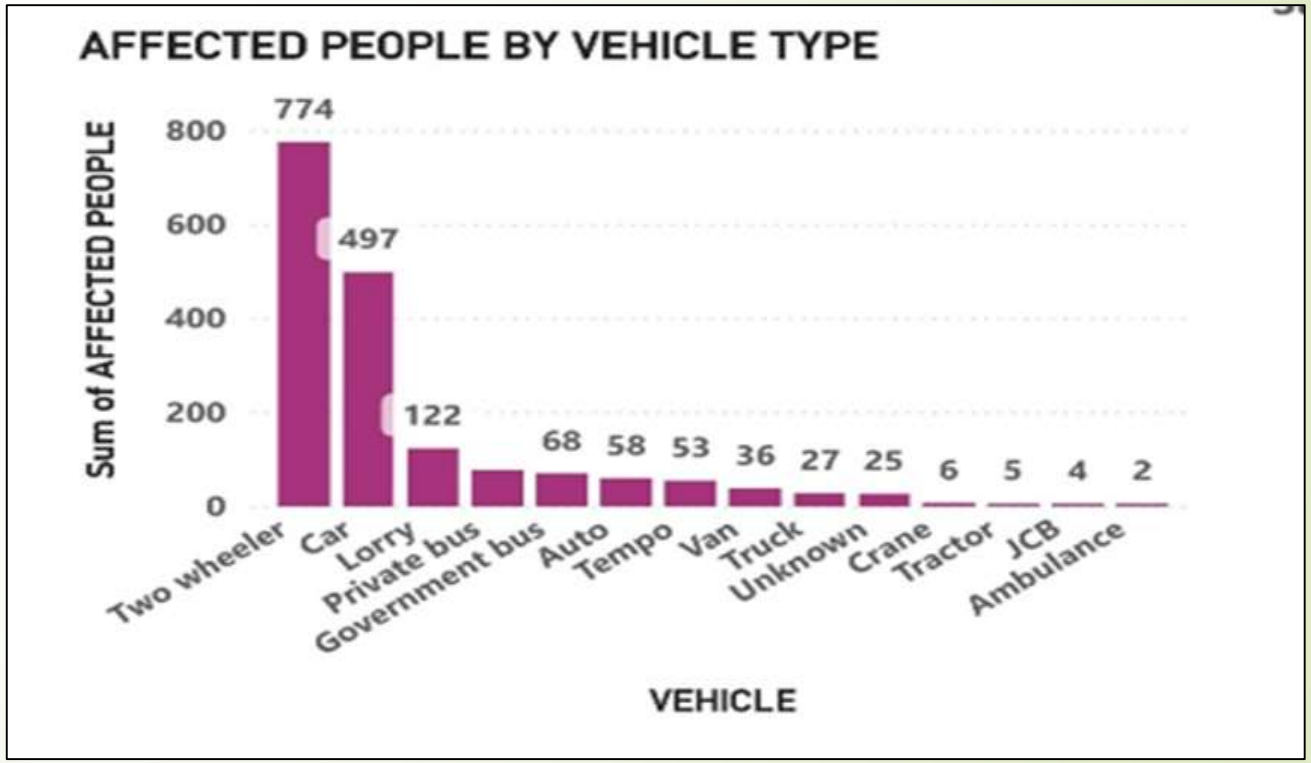
AIM: Evaluate the severity of accidents, including the number of injuries and fatalities, to understand the impact on public safety.

TOOLS USED: Chi square test (SPSS) , Visualization (Bar Graph in Power BI)

RESULTS:

Chi-Square Tests			
	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	207.485[a]	104	.000
Likelihood Ratio	103.301	104	.501
N of Valid Cases	1546		
107 cells (84.9%) have expected count less than 5. The minimum expected count is .00.			

Chi-Square test



Bar Graph for Affected People by Vehicle Type

INTERPRETATION:

- In the chi-square test, since the p- value (0.000) is lesser than level of significance 0.05, we may reject the null hypothesis Ho. Hence, we conclude that there is **significant association** between Vehicle and Affected People.
- Using Power BI , we found that the bar chat illustrating affected people by vehicle type shows that **two-wheelers** account for the highest number of accidents compared to other vehicle types.

4) Evaluate Demographic Factors:

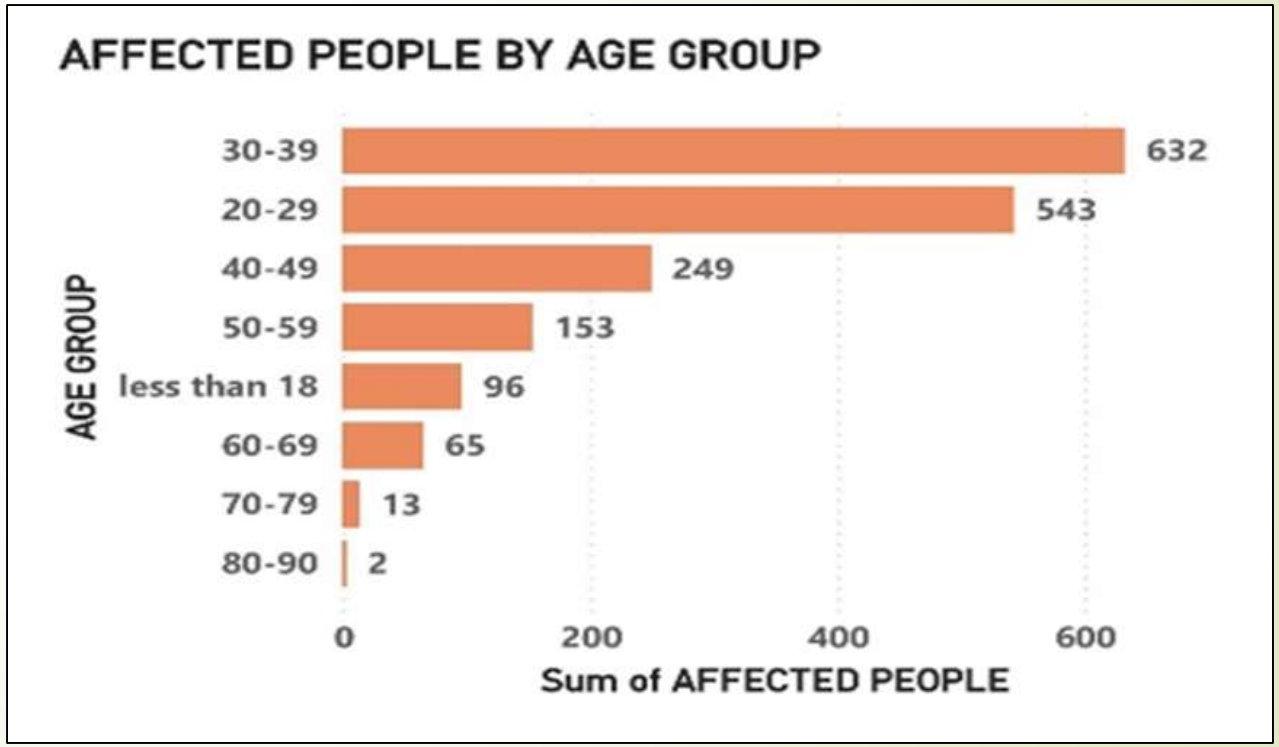
AIM: Study the involvement of different demographic groups, such as age, gender, and occupation, in accidents.

TOOLS USED : Chi square test (SPSS) , Visualization (Stacked Bar Chart in Power BI)

RESULTS:

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	537.607[a]	504	.145
Likelihood Ratio	249.308	504	1.000
Linear-by-Linear Association	.058	1	.810
N of Valid Cases	1546		
524 cells (91.0%) have expected count less than 5. The minimum expected count is .00.			

Chi-Square test



Stacked Bar Chart for Affected People by Age Group

INTERPRETATION:

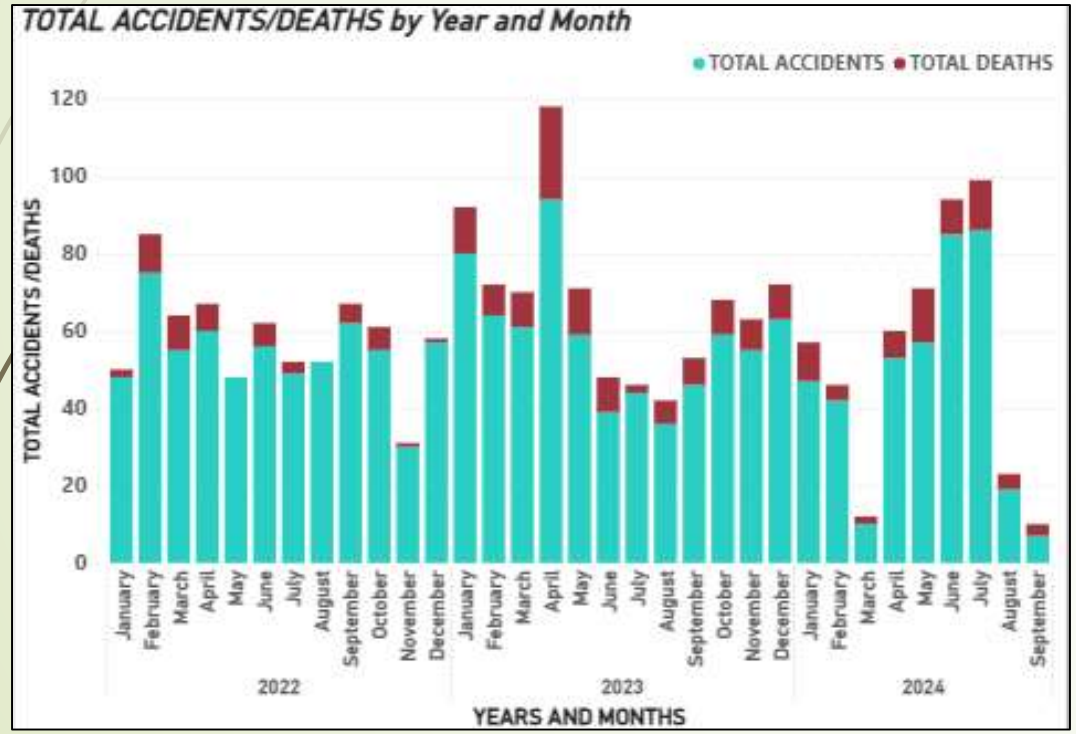
- In the chi square test , since the p- value (0.145) is greater than level of significance 0.05, we may accept the null hypothesis Ho . Hence, we conclude that there is **no significant association** between Age Group and Affected People.
- The stacked bar chart shows the number of accidents by age group reveals that individuals **between 30-39** years are most frequently involved, with this group showing the highest number of affected people compared to other age ranges.

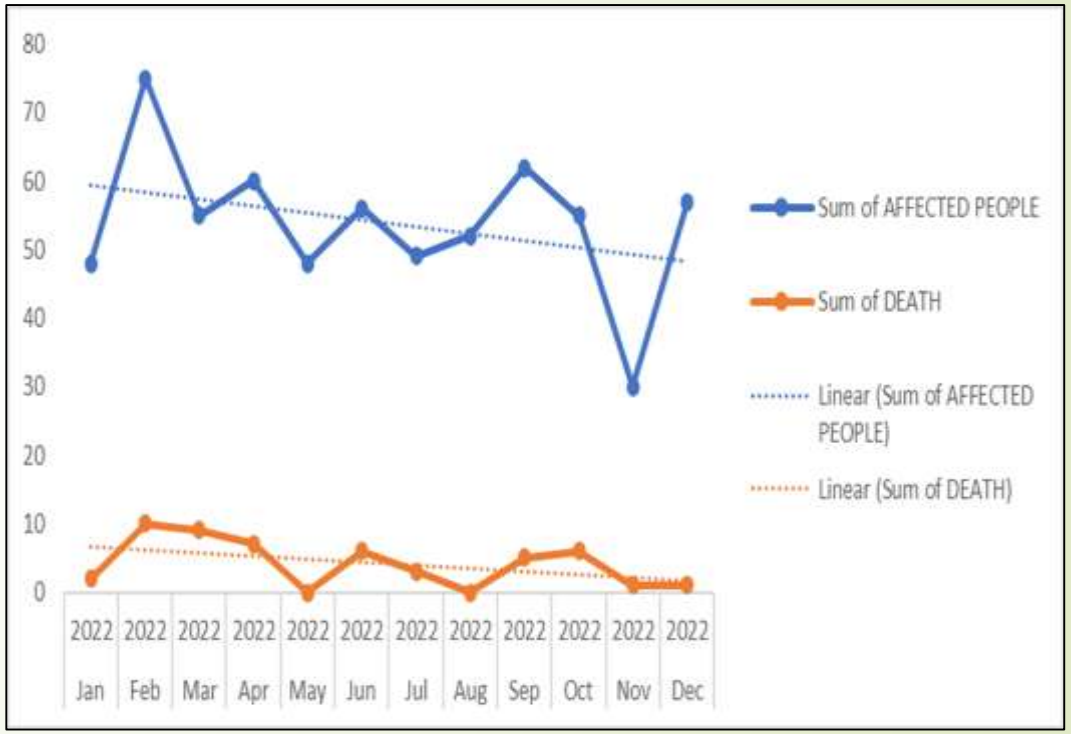
5) Understand Temporal Patterns:

AIM: Analyze the timing of accidents to identify peak times or seasons when accidents are more likely to occur.

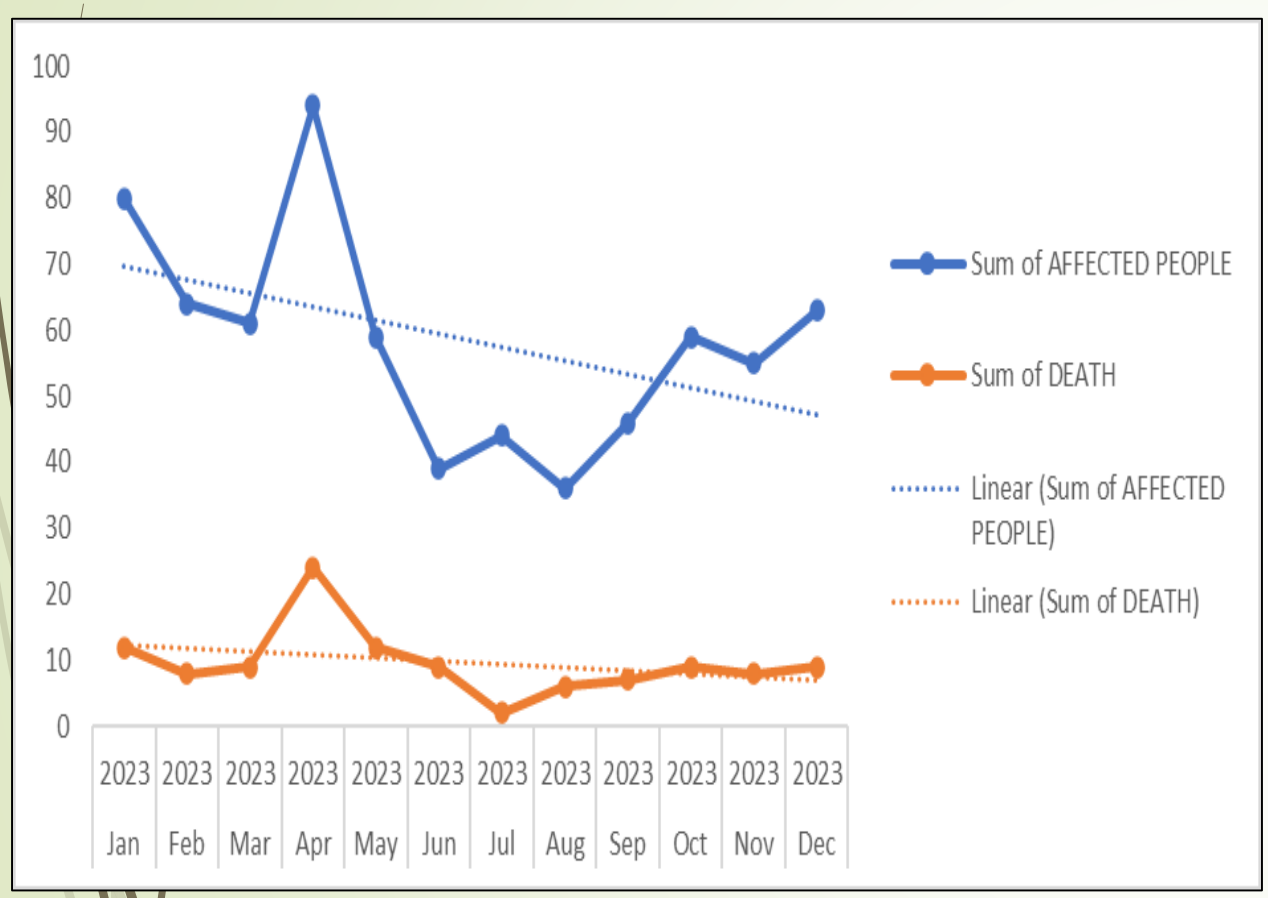
TOOLS USED: Trend line chart (Excel),Power BI

RESULTS:

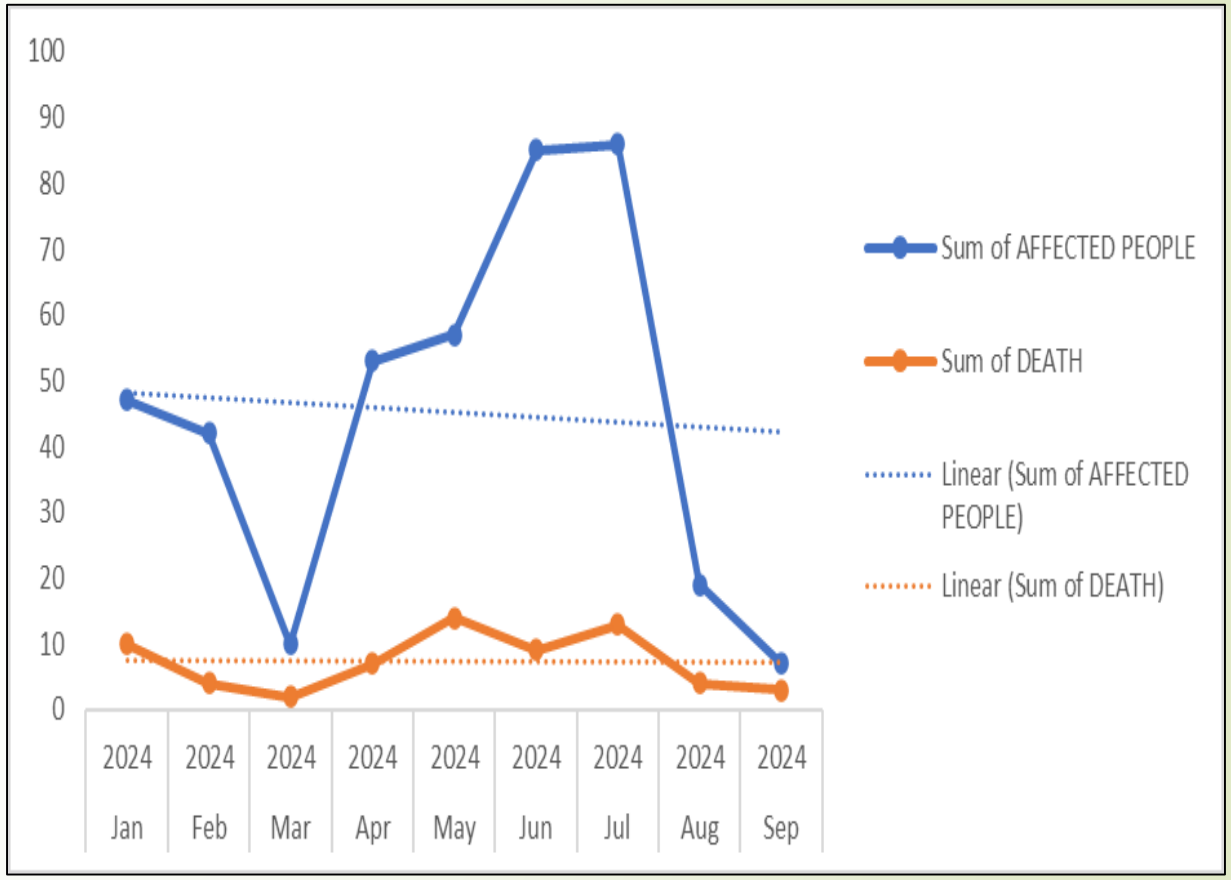




Line chart for 2022 using Trend line



Line chart for 2023 using Trend line



Line chart for 2024 using Trend line

INTERPRETATION:

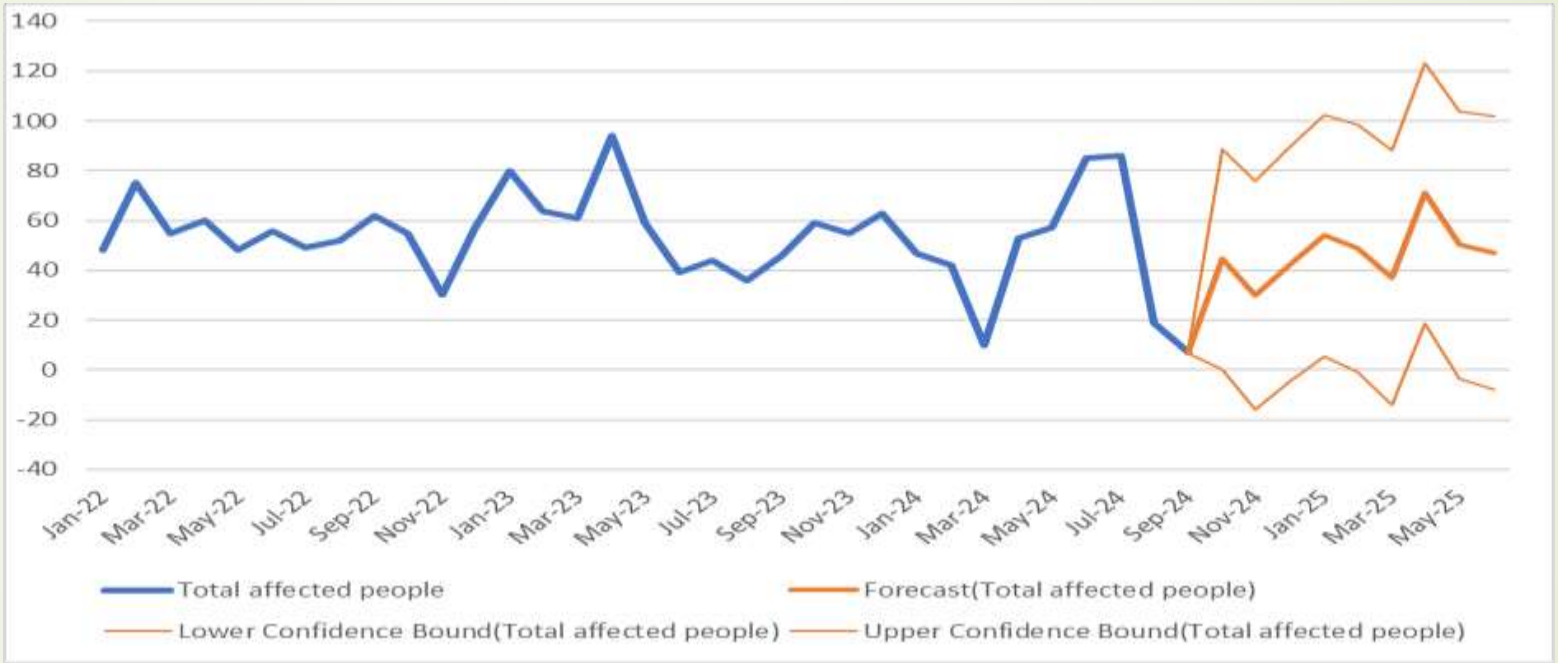
- For the year 2022,**February** experienced the highest number of accidents, with a notable spike in both the sum of affected individuals and the sum of deaths compared to other months of the year and also the **trend line goes downwards**.
- For the year 2023, **April** recorded the highest number of accidents compared to other months. The sum of affected individuals reached its peak during this month, and there was also a **noticeable increase** in the sum of deaths, indicating that these accidents were more severe and the trend line shows a **downward tendency**
- For the year 2024 (up to September), the months of **June and July** saw a marked increase in the number of accidents, with both months recording higher figures in terms of the sum of affected individuals and the sum of deaths compared to other months of the year.

6) Forecasting :

AIM : To predict the number of accidents that happen in the future

TOOLS USED: Forecast sheet (Excel)

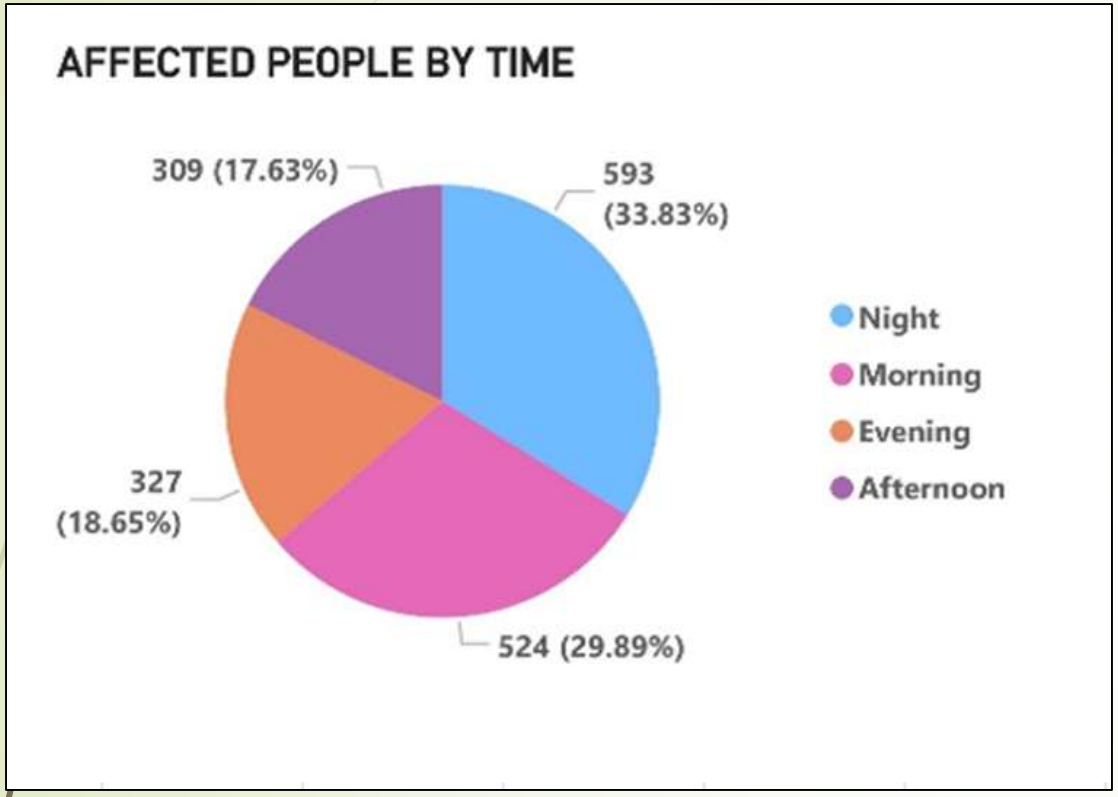
RESULTS:



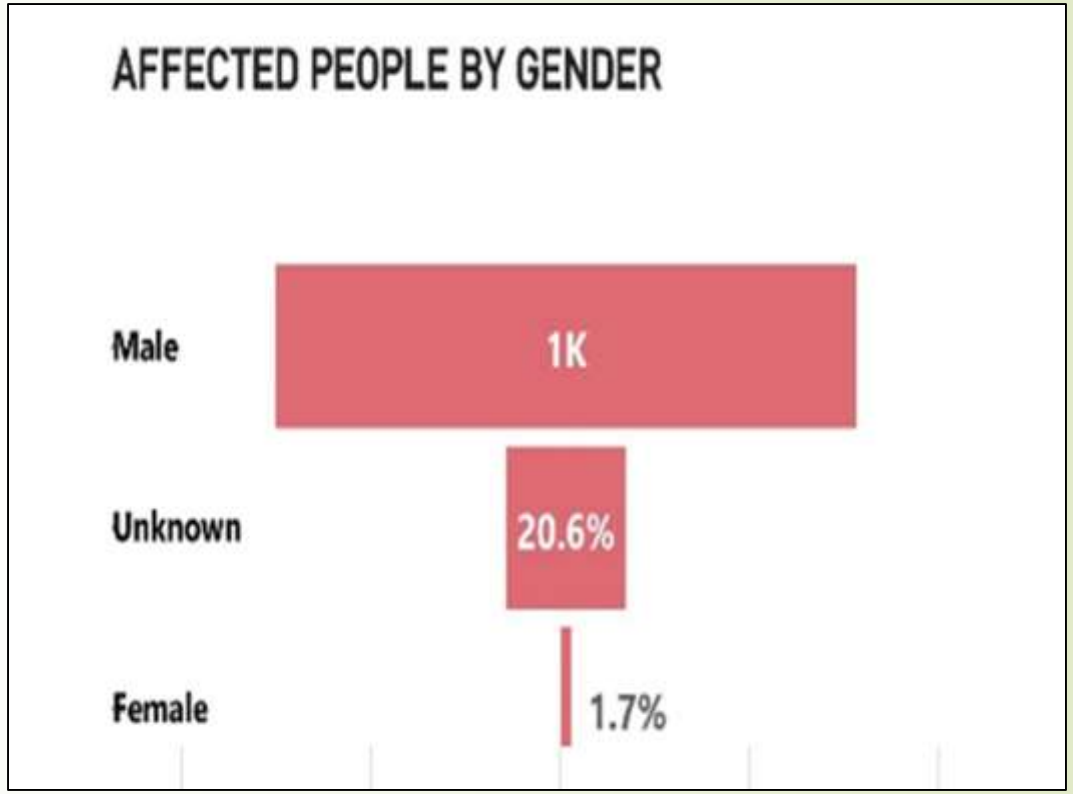
Forecasted graph

INTERPRETATION: From the above forecast chart, we could say that the highest number of accidents that could happen is during the month of **April 2025(**approx. 71) and the lowest number of accidents may be during the month of **November 2024 (**approx. 31).

OTHER RESULTS:



Pie chart for Affected People by Time



Funnel Chart for Affected People by Gender

CONCLUSION

- ***Rash and negligent driving or drunk and drive,*** result in a high number of accidents.
- The ***high*** standard deviation (***252.32***) and variance further indicate that the number of accidents caused by different factors ***varies widely***.
- ***February*** (75) and ***November*** (30) has the most and least number of accidents in the year of 2022 respectively.
- ***April*** (94) and ***August*** (36) has the most and least number of accidents in the year of 2023 respectively.
- ***July*** (86) and ***March*** (10) has the most and least number of accidents in the year of 2024 up to the month of September respectively.
- Individuals between ***30-39 years (632)*** are most frequently involved in accidents.

- The number of accidents by time of day reveals that ***night time*** accounts for the highest number of accidents.
- ***Males*** are involved in a significantly higher number of accidents compared to females.
- Also, ***Two-wheelers*** account for the highest number of accidents compared to other vehicle types.
- There is ***significant association*** between Vehicle and Age group with the affected people.
- The highest number of accidents took place in the ***Avinashi Road (232)*** and the least number of accidents took place in the ***Vilankurichi Road (29).***
- We predicted the number of accidents that could happen from the month of Oct 2024 till June 2025.
- The prediction concluded that ***April 2025*** (71) will have a greater number of accidents compared with other months.

SUGGESTION

🚗 **Strict enforcement**

1. Increase traffic police ***patrols*** on major accident-prone roads.
2. Use ***AI-based speed cameras and red-light cameras***.

📚 **Driver awareness programs**

1. Mandatory ***road safety training*** in colleges and workplaces.
2. Campaigns focusing on:
   - Speed control
   - Lane discipline
   - Helmet and seatbelt usage

📱 **Distracted driving control**

Penalize mobile phone usage while driving.

🎓 **Target young drivers**

1. Road safety ***workshops*** in IT parks, colleges, and industries.
2. Incentivize safe driving programs in companies.

🚦 **Driving behavior monitoring**

Encourage use of ***telematics / safe-driving apps.***

⛑ **Helmet enforcement**

Strict helmet checks for riders and pillion passengers.

🚦 **Better road markings**

Clear lane separation for bikes.

🔦 **Improve infrastructure**

1. Better lighting
2. Speed calming measures
3. Pedestrian crossings
4. Smart signals

Road Accident Analysis – Coimbatore (2022–24)

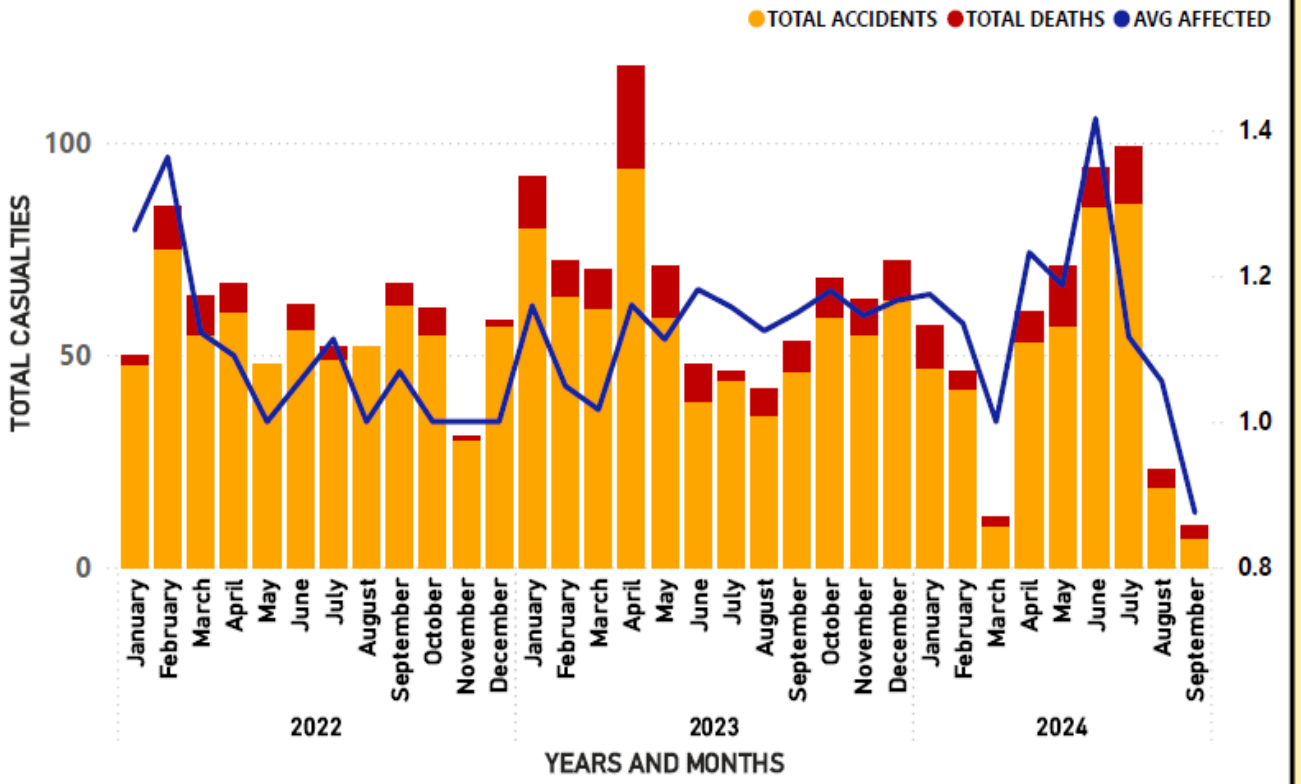
2022 | 2023 | 2024

1753	231	1554	13.18
TOTAL ACCIDENTS	TOTAL DEATHS	TOTAL AFFECTED PEOPLE	FATALITY RATE (%)

DAYS: All

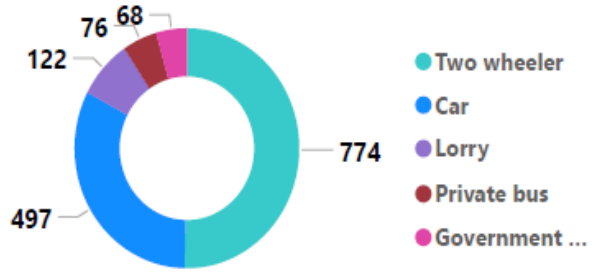
MONTHS: All

ACCIDENT TREND OVER THE YEARS



CAUSE	TOTAL ACCIDENTS	TOTAL DEATHS
Rash and negligent driving	1643	212
Self hit	28	13
Drunk and drive	74	6
Hit and run	8	0

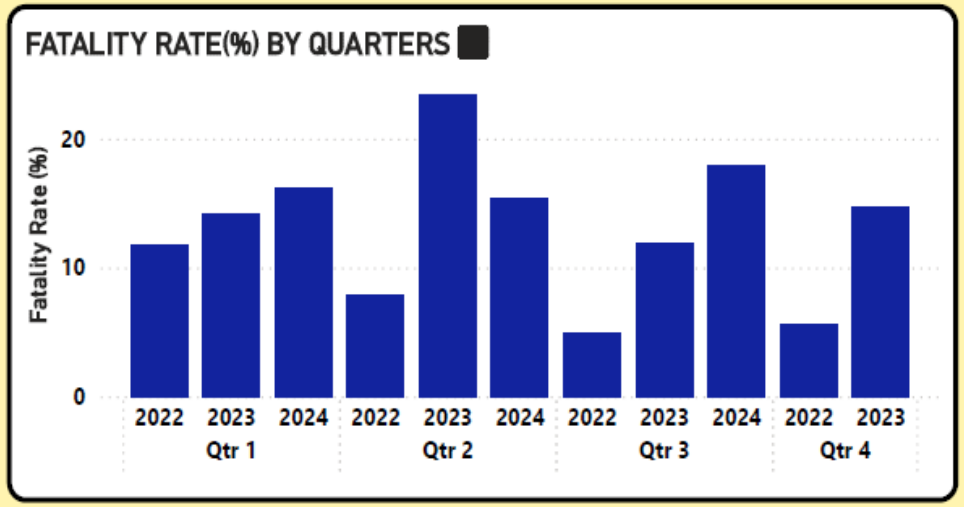
TOTAL ACCIDENTS BY VEHICLE

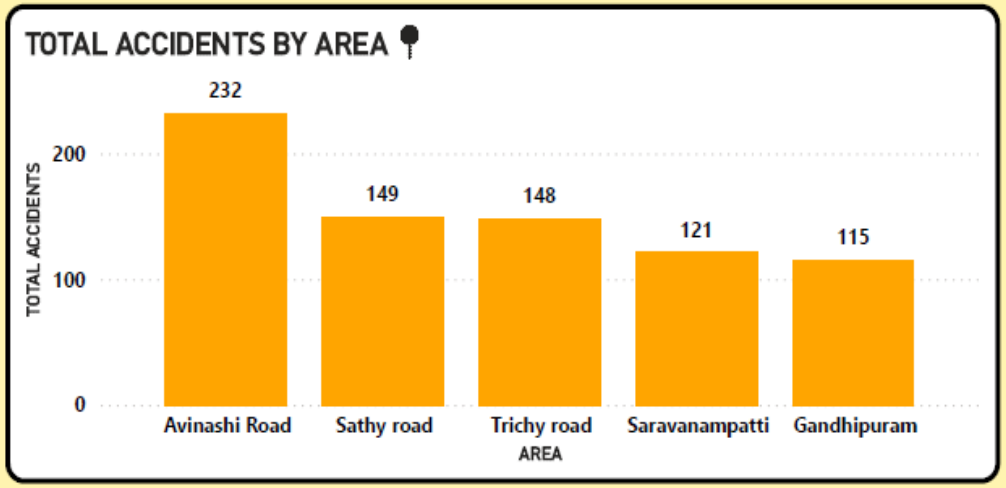


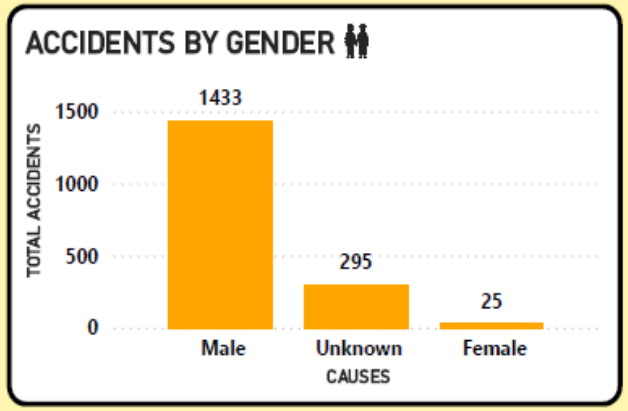
Road Accident Analysis – Coimbatore (2022–24)

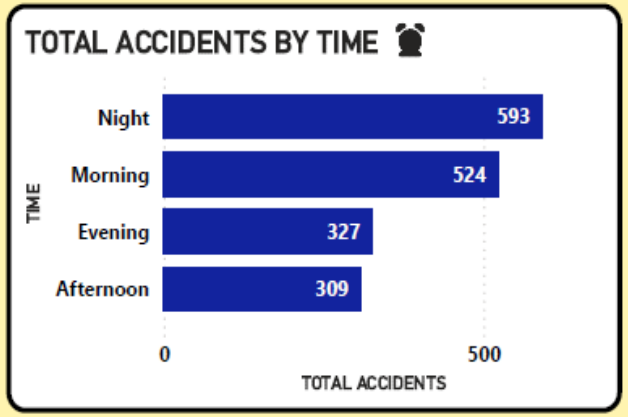
1753
TOTAL ACCIDENTS

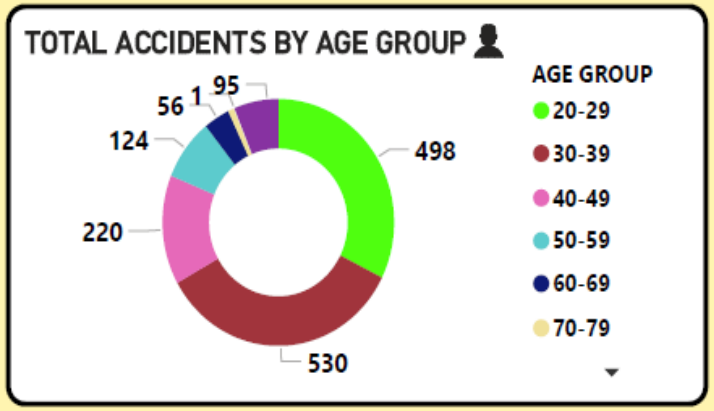
231
TOTAL DEATHS











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THANK YOU !!

-VIKASINI D

(vikasinidevaraj.08@gmail.com)